

A Casimir Ladder for the Electroweak Bosons: Exact Algebra of the de Vries–Rivero Ratio, a Boson Tower below 107 GeV, and Falsifiable Consequences

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Abstract

We study the two-parameter algebraic structure underlying a family of empirical electroweak mass relations first observed by H. de Vries and A. Rivero in 2004–2006 and since investigated, by several independent routes, in a corpus of research programs. The structure is a one-line kernel: to an internal label j with Casimir $J = j(j+1)$ associate the positive root $\lambda_+(J)$ of $\lambda^2 + J\lambda - J = 0$, and set $m^2(j) = M_\infty^2 \lambda_+(J)$ for a single scale M_∞ . With the channel assignment $(\gamma, W, Z) = (0, \frac{1}{2}, 1)$ this gives $m_W/m_Z = \sqrt{\lambda_+(3/4)/\lambda_+(2)} = 0.881419\dots$, within 0.4σ of the 2024 on-shell world average. We prove three structural results. (i) The kernel is the unique attractor of a nested Feshbach decimation with uniform rung data $(\sqrt{J}, J)$; we also record an obstruction the source literature passes over — a *flat* semi-infinite chain does not work, since the candidate boundary state is an embedded resonance for both physical Casimirs. (ii) The mixing parameter implied by the kernel, $\sin^2 \theta_W = 1 - \lambda_+(3/4)/\lambda_+(2) = (19 - 3\sqrt{19} + 3\sqrt{3} - \sqrt{57})/16 = 0.2231013\dots$, has minimal polynomial $64x^4 - 304x^3 + 414x^2 - 190x + 25$ over $\mathbb{Q}$: this is precisely the quartic postulated independently in a second research line, so two formulations believed merely numerically close are algebraically identical; moreover the quartic’s four roots are exactly the four branch combinations $1 - \lambda_a(J_W)/\lambda_b(J_Z)$, and the polynomial embeds in an explicit two-parameter family whose coefficients are polynomial in the channel Casimir and the ratio $\kappa = J_W/J_Z = 3/8$ isolated independently by a custodial-EFT analysis. (iii) The negative branch obeys two exact sum rules, and the popular identifications of its levels with $v/2$ and $v/\sqrt{2}$ are mutually inconsistent at the 1.8% level, so at most one can be exact. The positive branch predicts a tower of states at 96.5, 99.6, 101.5, 102.7 GeV accumulating at $M_\infty = m_Z \sqrt{(1 + \sqrt{3})/2} = 106.58$ GeV, with a label-conservation pattern that suppresses single production of the $j = 3/2$ level — compatible with the weak diphoton hints near 95–96 GeV and decisively testable at FCC-ee, with LEP 2 recasts above the 193 GeV pair threshold available already. We state which assumptions remain underived, give the failure conditions, and enumerate the trials factors any significance claim must price.

1 Introduction

The ratio of the W and Z boson masses is, in the Standard Model (SM), a derived quantity: $m_W/m_Z = \cos \theta_W$ at tree level, with θ_W fixed by the ratio of two independent gauge couplings.

*This manuscript was produced by a language model as a demonstration of autonomous theoretical synthesis, under an explicit constraint of zero code execution: every closed form was derived symbolically and every decimal obtained by hand arithmetic from those closed forms. The source material is a corpus of research repositories maintained by A. Rivero; Section 11 states which results are inherited, which are reproved, and which are new. The bibliography was verified by a second model (Claude Sonnet) against arXiv and the published record; an adversarial referee report was solicited from an independent model (GPT-5.5, via the Codex CLI) on a first draft, and this version incorporates the resulting corrections — including one genuine algebra error it caught. Both reports are archived alongside the source.

The SM assigns it no preferred value. It is therefore legitimate — in the same exploratory spirit in which Koide’s charged-lepton relation has been studied for four decades [4, 5, 6, 7, 8, 9] — to ask whether the measured value singles out a simple algebraic structure, and, if one is found, to work out its full set of consequences so that data can kill it or promote it.

An observation of this kind has an arXiv trail going back two decades. In November 2004 H. de Vries noted [1] that the on-shell weak mixing parameter agrees closely with a specific algebraic number built from the two smallest nontrivial $SU(2)$ Casimirs; the observation was developed in a joint preprint with A. Rivero [2] and stated in kernel form — the eigenvalue problem used in this paper, with the value $s_{dV}^2 = 0.22310132\dots$ appearing verbatim — in [3]. A corpus of research repositories has since attacked the structure from several directions: as the secular equation of a two-channel kernel, as a root of an explicit quartic polynomial, as an eigenvalue problem for a Casimir-locked texture, and as a coefficient in a custodially symmetric effective field theory. The connections between these formulations had been checked numerically, never algebraically.

This paper does three things.

First, it states the structure once, minimally, and derives its exact spectrum in closed form (Sections 2, 3). The kernel is a one-parameter family of 2×2 matrices; we prove it is equivalently the unique attractor of a nested decimation recursion with uniform rung data (Proposition 1), and we record an obstruction that the source literature passes over: a flat semi-infinite chain has the candidate state as an embedded resonance rather than an eigenvalue (Remark 1), which constrains what a microscopic origin can look like.

Second, it proves the algebraic equivalence of the formulations (Section 4). Theorem 1 computes, via Galois conjugation in the biquadratic field $\mathbb{Q}(\sqrt{3}, \sqrt{19})$, the minimal polynomial of the mixing parameter implied by the kernel, and finds exactly the quartic $64x^4 - 304x^3 + 414x^2 - 190x + 25$ that a separate research line had postulated as its fundamental object. What was believed to be two compatible proposals is one proposal.

Third, it extracts the falsifiable consequences derivable without new dynamical input (Sections 5–7): a 2×10^{-5} -precision prediction for m_W from m_Z (at tree level, on-shell — the scheme caveat is discussed at length); a tower of states between 96.5 and 106.6 GeV with computable asymptotic spacing; a label-conservation pattern suppressing their single production; two exact sum rules for the negative branch; and a no-go (Lemma 1) showing that the two popular vacuum-scale identifications of the negative branch cannot both be exact.

Section 8 records a further coincidence: the coefficient $\kappa = 3/8 = C_F/C_A$ of $SU(2)$, isolated independently by a custodial-EFT analysis of the same data, is the ratio J_W/J_Z of the very Casimirs that enter the kernel. Section 9 discusses which geometric mechanisms could produce the kernel [13, 14, 15, 16, 19, 18, 17] and states, as numbered assumptions, exactly what a derivation would have to produce. Section 10 collects the failure conditions, and Section 11 the provenance, method, and the trials accounting that any significance claim must confront.

Throughout, “prediction” means: output of the closed formulas given here, with stated inputs, at tree level, compared in the on-shell scheme. The scheme question is the single largest interpretive uncertainty and is treated in Section 5.

2 The kernel and the decimation recursion

2.1 Definition

Fix a mass scale M_∞ . To an internal label $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ with Casimir $J = j(j+1)$ associate the matrix

$$A(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (1)$$

acting on a two-dimensional space spanned by a “visible” level (first slot) and a “partner” level (second slot), in units of M_∞^2 . Its characteristic equation is

$$\lambda^2 + J\lambda - J = 0, \quad \lambda_\pm(J) = \frac{-J \pm \sqrt{J^2 + 4J}}{2}, \quad (2)$$

with $\lambda_+(J) \in (0, 1)$ and $\lambda_-(J) < 0$ for all $J > 0$. The physical mass-squared assignments studied here are

$$m^2(j) = M_\infty^2 \lambda_+(j(j+1)) \quad (\text{positive branch}), \quad (3)$$

with the channel assignment

$$\gamma \leftrightarrow j = 0, \quad W \leftrightarrow j = \frac{1}{2}, \quad Z \leftrightarrow j = 1, \quad (4)$$

i.e. $J_\gamma = 0$ (massless), $J_W = \frac{3}{4}$, $J_Z = 2$. The label j is distinct from the gauged weak isospin of the SM (under which $W^\pm, Z$ form a triplet); it is a bookkeeping label of the construction, tentatively interpreted as a representation label of an internal (custodial-like) $SU(2)_i$, and assignment (4) is an *input*, inherited from [3], whose derivation is an open problem. We return to its status in Sections 9 and 11.

Two elementary properties of (2) are used repeatedly:

$$\lambda_+ + \lambda_- = -J, \quad \lambda_+ \lambda_- = -J \quad \implies \quad |\lambda_-| - \lambda_+ = J, \quad \lambda_+ |\lambda_-| = J, \quad (5)$$

together with the rewriting $\lambda_+^2 = J(1 - \lambda_+)$ of (2) itself.

2.2 The resummation identity, and what it is not

The matrix (1) looks like a fine-tuned two-state system. The following identity shows the same spectrum arises as the fixed point of a self-similar decimation, which widens the space of candidate microscopic origins.

Proposition 1 (Resummation identity). *Let $f(X) = J/(J+X)$ on $[0, \infty)$, for fixed $J > 0$. Then:*

- (i) *the fixed points of f are exactly the roots $\lambda_\pm(J)$ of (2);*
- (ii) *for every $X_0 \geq 0$ the iterates $X_{k+1} = f(X_k)$ converge to $\lambda_+(J)$;*
- (iii) *f is the static (zero-frequency) Schur reduction of the two-level block $\begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -(J+X) \end{pmatrix}$ onto its visible slot: eliminating a partner that carries, besides the Casimir mass $-J$, the effective mass $-X$ handed up from the stage below, returns the effective visible mass $f(X)$. Hence any nested decimation scheme with uniform rung data $(\sqrt{J}, J)$ has $\lambda_+(J)$ as its unique attractor.*

Proof. (i) $f(X) = X$ is $X(J+X) = J$, i.e. (2). (ii) f is a Möbius transformation mapping $[0, \infty)$ into $(0, 1]$, strictly decreasing, with $f([0, 1]) = [J/(J+1), 1]$. Hence $f \circ f$ is increasing, so the even and odd subsequences of $(X_k)_{k \geq 1}$ are monotone and confined to $[J/(J+1), 1]$; each converges to a fixed point of $f \circ f$. Now $f \circ f$ is itself a Möbius transformation and is distinct from the identity (e.g. $f(f(0)) = J/(J+1) \neq 0$), so it has at most two fixed points; the two fixed points $\lambda_\pm$ of f already account for them, leaving no room for genuine 2-cycles. Both subsequence limits lie in $(0, 1]$ while $\lambda_- < 0$, so both equal λ_+ . (For the record, the multiplier is $|f'(\lambda_+)| = \lambda_+^2/J = 1 - \lambda_+ < 1$, using $\lambda_+^2 = J(1 - \lambda_+)$.) (iii) The Schur complement of the stated block onto the first slot at zero frequency is $0 - \sqrt{J}(-(J+X))^{-1}\sqrt{J} = J/(J+X) = f(X)$. $\square$

Remark 1 (What the ladder is not). It is tempting to read Proposition 1 as the statement that a *flat* semi-infinite chain — the tridiagonal operator T with diagonal $(0, -J, -J, \dots)$ and off-diagonals $\sqrt{J}$ — possesses λ_+ as a boundary-localized eigenvalue. It does not. The tail of

T has essential spectrum $[-J - 2\sqrt{J}, -J + 2\sqrt{J}]$, and λ_+ lies inside this band precisely when $(\lambda_+ + J)^2 < 4J$, which by $\lambda_+ + J = J/\lambda_+$ and $\lambda_+^2 = J(1 - \lambda_+)$ reduces to $\lambda_+ < \frac{3}{4}$, i.e. $J < \frac{9}{4}$. Both physical values $J_W = \frac{3}{4}$ ($\lambda_+ = 0.5687$) and $J_Z = 2$ ($\lambda_+ = 0.7321$) satisfy this, so on the flat chain the candidate boundary state is an embedded resonance rather than an isolated eigenvalue. The correct reading of Proposition 1 is therefore recursive-hierarchical, as in item (iii): each partner carries the effective mass of the stage below. Microscopic proposals must implement a nested decimation — successive scales of a Kaluza–Klein-type recursion are the natural candidate — rather than a flat chain. We state this obstruction explicitly because the source corpus passes over it.

3 Exact spectrum and the number field

Evaluating (2) at the two physical Casimirs:

$$\lambda_+(\tfrac{3}{4}) = \frac{\sqrt{57} - 3}{8} = 0.568729\dots, \quad \lambda_-(\tfrac{3}{4}) = -\frac{3 + \sqrt{57}}{8} = -1.318729\dots, \quad (6)$$

$$\lambda_+(2) = \sqrt{3} - 1 = 0.732051\dots, \quad \lambda_-(2) = -(1 + \sqrt{3}) = -2.732051\dots \quad (7)$$

The mass-squared ratio on the positive branch is

$$\rho \equiv \frac{m_W^2}{m_Z^2} = \frac{\lambda_+(3/4)}{\lambda_+(2)} = \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)} = \frac{3\sqrt{19} + \sqrt{57} - 3\sqrt{3} - 3}{16} = 0.776899\dots, \quad (8)$$

where we rationalized with $(\sqrt{3} - 1)^{-1} = (\sqrt{3} + 1)/2$ and used $\sqrt{57} = \sqrt{3}\sqrt{19}$. Hence

$$\frac{m_W}{m_Z} = \sqrt{\rho} = 0.8814186\dots, \quad \sin^2 \theta_W \equiv 1 - \rho = \frac{19 - 3\sqrt{19} + 3\sqrt{3} - \sqrt{57}}{16} = 0.2231013\dots \quad (9)$$

All these numbers live in the real biquadratic field $K = \mathbb{Q}(\sqrt{3}, \sqrt{19})$, of degree 4 over $\mathbb{Q}$, with Galois group $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ generated by

$$\sigma : \sqrt{3} \mapsto -\sqrt{3}, \quad \tau : \sqrt{19} \mapsto -\sqrt{19},$$

each of which flips $\sqrt{57}$. A curiosity worth recording: since K is biquadratic, every quantity in this construction is expressible by nested square roots alone — in this model $\sin^2 \theta_W$ is a ruler-and-compass constructible number.

4 The equivalence theorem

A second research line in the corpus posits, as its fundamental object, the quartic polynomial

$$P(x) = 64x^4 - 304x^3 + 414x^2 - 190x + 25, \quad (10)$$

one of whose roots is matched against electroweak data. The two lines were believed compatible because their decimal expansions agree. We now prove they are the same object.

Theorem 1 (The quartic is the minimal polynomial of the mixing parameter). *Let $s = \sin^2 \theta_W = (19 - 3\sqrt{19} + 3\sqrt{3} - \sqrt{57})/16$ as in (9). Then the minimal polynomial of s over $\mathbb{Q}$ is exactly $P(x)$ of (10), normalized to leading coefficient 64. Consequently the kernel formulation (1) with channels (4) and the quartic formulation (10) define the same algebraic number, and the two research lines study one structure.*

Proof. The four Galois images of s under $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\}$ are obtained by the sign choices $\varepsilon_3, \varepsilon_{19} \in \{\pm 1\}$ on $\sqrt{3}$ and $\sqrt{19}$ (the sign of $\sqrt{57}$ being $\varepsilon_3\varepsilon_{19}$):

$$s(\varepsilon_3, \varepsilon_{19}) = \frac{19 - 3\varepsilon_{19}\sqrt{19} + 3\varepsilon_3\sqrt{3} - \varepsilon_3\varepsilon_{19}\sqrt{57}}{16}.$$

Write $s_1 = s(+, +) = s$, $s_2 = s(-, +)$, $s_3 = s(+, -)$, $s_4 = s(-, -)$.

The orbit has size four. Since $\{1, \sqrt{3}, \sqrt{19}, \sqrt{57}\}$ is a $\mathbb{Q}$ -basis of K , two images $s(\varepsilon)$, $s(\varepsilon')$ coincide only if their coefficient vectors coincide, i.e. only if $\varepsilon = \varepsilon'$; the four images are pairwise distinct. (Numerically: $s_1 = 0.223101$, $s_2 = 0.517311$, $s_4 = 1.208169$, $s_3 = 2.801418$.) Hence the stabilizer of s in the Galois group is trivial, $[\mathbb{Q}(s) : \mathbb{Q}] = 4$, and the degree-4 monic polynomial $\prod_i (x - s_i)$, which is Galois-invariant and therefore has rational coefficients, is irreducible over $\mathbb{Q}$ and is the minimal polynomial of s .

Computation of the coefficients. We compute the elementary symmetric functions by grouping conjugate pairs.

Sum: the $\sqrt{3}, \sqrt{19}, \sqrt{57}$ terms cancel over the four sign patterns, so $e_1 = 4 \cdot 19/16 = 19/4$.

Pair data at fixed ε_{19} : with $A = 19 - 3\varepsilon_{19}\sqrt{19}$ and $B = 3\sqrt{3} - \varepsilon_{19}\sqrt{57}$, the two values of ε_3 give $(A \pm B)/16$, so

$$s(+, \varepsilon_{19}) s(-, \varepsilon_{19}) = \frac{A^2 - B^2}{256}, \quad A^2 = 532 - 114\varepsilon_{19}\sqrt{19}, \quad B^2 = 84 - 18\varepsilon_{19}\sqrt{19},$$

using $(3\sqrt{3})^2 = 27$, $(\sqrt{57})^2 = 57$ and $2 \cdot 3\sqrt{3} \cdot \sqrt{57} = 6\sqrt{171} = 18\sqrt{19}$. Thus

$$s_1 s_2 = \frac{28 - 6\sqrt{19}}{16}, \quad s_3 s_4 = \frac{28 + 6\sqrt{19}}{16}, \quad s_1 + s_2 = \frac{19 - 3\sqrt{19}}{8}, \quad s_3 + s_4 = \frac{19 + 3\sqrt{19}}{8}.$$

$$\text{Constant term: } e_4 = s_1 s_2 s_3 s_4 = \frac{28^2 - 36 \cdot 19}{256} = \frac{784 - 684}{256} = \frac{25}{64}.$$

Second symmetric function:

$$e_2 = s_1 s_2 + s_3 s_4 + (s_1 + s_2)(s_3 + s_4) = \frac{56}{16} + \frac{361 - 171}{64} = \frac{7}{2} + \frac{190}{64} = \frac{207}{32}.$$

Third symmetric function:

$$e_3 = s_1 s_2 (s_3 + s_4) + s_3 s_4 (s_1 + s_2) = \frac{(28 - 6\sqrt{19})(19 + 3\sqrt{19}) + (28 + 6\sqrt{19})(19 - 3\sqrt{19})}{128} = \frac{(190 - 30\sqrt{19}) + (190 + 30\sqrt{19})}{128}$$

Therefore the minimal polynomial is

$$x^4 - \frac{19}{4}x^3 + \frac{207}{32}x^2 - \frac{95}{32}x + \frac{25}{64} = \frac{1}{64}P(x). \quad \square$$

Corollary 1. (a) The mass-squared ratio $\rho = 1 - s$ has minimal polynomial $64y^4 + 48y^3 - 114y^2 + 18y + 9$, obtained by expanding $P(1 - y)$. (b) The mass ratio $z = m_W/m_Z = \sqrt{\rho}$ has degree 8 over $\mathbb{Q}$, with minimal polynomial $64z^8 + 48z^6 - 114z^4 + 18z^2 + 9$.

Proof. (a) Direct substitution; the coefficients follow from the binomial expansions of $(1 - y)^k$, $k \leq 4$. (b) $z^2 = \rho$ with $[\mathbb{Q}(\rho) : \mathbb{Q}] = 4$, so $[\mathbb{Q}(z) : \mathbb{Q}] \in \{4, 8\}$, and it is 8 unless ρ is a square in $K = \mathbb{Q}(\rho)$. Suppose $\rho = \beta^2$ with $\beta \in K$. K is a subfield of $\mathbb{R}$ and Galois over $\mathbb{Q}$, so every Galois image of ρ would be the square of a real number, hence nonnegative. But the image of $\rho = 1 - s$ under τ is $1 - s_3 = -1.801418 \dots < 0$. Contradiction; so $[\mathbb{Q}(z) : \mathbb{Q}] = 8$ and the stated monic-normalized degree-8 polynomial, of which z is a root, is its minimal polynomial. $\square$

Remark 2. Theorem 1 is a statement about the internal consistency of the theoretical corpus, and it *removes* one apparent independent confirmation: the quartic line and the kernel line are one coincidence, so the naive over-determination count goes down even as the structural economy goes up. The bookkeeping consequences are drawn in Section 11.

4.1 The branch structure of the quartic: a two-parameter family

The proof of Theorem 1 contains more information than the statement. Tracking the Galois action on the closed forms (6)–(7): τ (which flips $\sqrt{57}$ but fixes $\sqrt{3}$) swaps $\lambda_+(\frac{3}{4}) \leftrightarrow \lambda_-(\frac{3}{4})$ while fixing $\lambda_\pm(2)$; $\sigma\tau$ swaps $\lambda_+(2) \leftrightarrow \lambda_-(2)$ while fixing $\lambda_\pm(\frac{3}{4})$; and σ swaps both. Hence the four roots of the corpus quartic are precisely the four *branch combinations*

$$s_{ab} = 1 - \frac{\lambda_a(J_W)}{\lambda_b(J_Z)}, \quad a, b \in \{+, -\} :$$

$s_{++} = \sin^2 \theta_W = 0.223101$, $s_{--} = 0.517311$, $s_{+-} = 1.208169$, $s_{-+} = 2.801418$, and the Galois group of Theorem 1 acts as the group of branch swaps. The “spurious” roots of the quartic are the other branch choices — the arithmetic of the polynomial is the physics of the branch structure. This observation globalizes:

Theorem 2 (Two-parameter branch quartic). *Fix channel Casimirs $I, K > 0$ and let $u_\pm = \lambda_\pm(I)$, $w_\pm = \lambda_\pm(K)$ be the roots of $p_I(x) = x^2 + Ix - I$ and $p_K(x) = x^2 + Kx - K$. Set $\kappa = I/K$ and $s_{ab} = 1 - u_a/w_b$. Then, identically in I, K ,*

$$\prod_{a,b \in \{\pm\}} (x - s_{ab}) = x^4 - (4 + I)x^3 + \left[6 + 2I - \kappa(I + 2)\right]x^2 - (4 + I)(1 - \kappa)x + (1 - \kappa)^2.$$

At $(I, K) = (J_W, J_Z) = (\frac{3}{4}, 2)$, i.e. $\kappa = \frac{3}{8}$, this is $P(x)/64$ with P the corpus quartic (10).

Proof. Vieta for the two channels gives $u_+ + u_- = -I$, $u_+ u_- = -I$, $w_+ + w_- = -K$, $w_+ w_- = -K$; hence the two identities used throughout: $\frac{1}{w_+} + \frac{1}{w_-} = \frac{w_+ + w_-}{w_+ w_-} = 1$ and $\frac{1}{w_+^2} + \frac{1}{w_-^2} = 1 + \frac{2}{K}$, together with $u_+^2 + u_-^2 = I^2 + 2I$. The key on-shell identity: if $w^2 + Kw - K = 0$ then $w^2 = K(1 - w)$, so

$$p_I(w) = w^2 + Iw - I = K - Kw + Iw - I = (K - I)(1 - w).$$

Constant term. Since $\prod_a (1 - u_a/w) = [w^2 - (u_+ + u_-)w + u_+ u_-]/w^2 = p_I(w)/w^2$,

$$e_4 = \prod_b \frac{p_I(w_b)}{w_b^2} = \frac{(K - I)^2 \prod_b (1 - w_b)}{(w_+ w_-)^2} = \frac{(K - I)^2 p_K(1)}{K^2} = \left(\frac{K - I}{K}\right)^2 = (1 - \kappa)^2,$$

using $p_K(1) = 1 + K - K = 1$. *Sum.* $e_1 = \sum_{ab} s_{ab} = 4 - (\sum_a u_a)(\sum_b 1/w_b) = 4 - (-I)(1) = 4 + I$.

Second symmetric function. From $\sum_{ab} s_{ab}^2 = 4 - 2(\sum u)(\sum 1/w) + (\sum u^2)(\sum 1/w^2) = 4 + 2I + (I^2 + 2I)(1 + \frac{2}{K})$ and $e_2 = \frac{1}{2}(e_1^2 - \sum s^2)$ one finds $e_2 = 6 + 2I - (I^2 + 2I)/K = 6 + 2I - \kappa(I + 2)$.

Third symmetric function. Using $e_3 = e_4 \sum_{ab} 1/s_{ab}$ and $1/s_{ab} = w_b/(w_b - u_a)$: for fixed $w = w_b$,

$$\sum_a \frac{w}{w - u_a} = \frac{w p'_I(w)}{p_I(w)} = \frac{w(2w + I)}{(K - I)(1 - w)} = \frac{(2K - I)(1 - w) + I}{(K - I)(1 - w)},$$

where the numerator was rewritten on-shell: $w(2w + I) = 2w^2 + Iw = 2K(1 - w) + Iw = 2K - (2K - I)w = (2K - I)(1 - w) + I$. Summing over b with $\sum_b 1/(1 - w_b) = p'_K(1)/p_K(1) = 2 + K$ gives

$$\sum_{ab} \frac{1}{s_{ab}} = \frac{2(2K - I) + I(2 + K)}{K - I} = \frac{K(4 + I)}{K - I}, \quad e_3 = (1 - \kappa)^2 \cdot \frac{K(4 + I)}{K - I} = (4 + I)(1 - \kappa).$$

The four coefficients assemble into the stated quartic. Specializing $I = \frac{3}{4}$, $\kappa = \frac{3}{8}$: $e_1 = \frac{19}{4}$, $e_2 = 6 + \frac{3}{2} - \frac{3}{8} \cdot \frac{11}{4} = \frac{207}{32}$, $e_3 = \frac{19}{4} \cdot \frac{5}{8} = \frac{95}{32}$, $e_4 = \frac{25}{64}$, matching Theorem 1. (As an independent check of the general identity, at the degenerate point $I = K = 2$ the formula gives roots $\{0, 0, 3 - \sqrt{3}, 3 + \sqrt{3}\}$ with $e_1 = 6$, $e_2 = 6$, $e_3 = e_4 = 0$, which is what the branch values $1 - \lambda_a/\lambda_b$ of a single channel indeed are.) $\square$

Remark 3. Theorem 2 demystifies the corpus quartic twice over. First, the integers (64, 304, 414, 190, 25) carry no independent information: they are the image of the two-parameter family at $(I, \kappa) = (\frac{3}{4}, \frac{3}{8})$, with 64 the least common denominator. Second, the coefficients depend on the pair (I, K) only through I and the ratio $\kappa = I/K$ — the same $\kappa = \frac{3}{8}$ isolated by the custodial-EFT line (Section 8), which therefore sits literally inside the quartic’s x^1 and x^0 coefficients, $e_3 = e_1(1 - \kappa)$ and $e_4 = (1 - \kappa)^2$. For the statistics program the family is equally useful: it *is* the natural pre-registerable hypothesis class (scan (I, K) over Casimir pairs) inside which the look-elsewhere cost of the choice $(\frac{3}{4}, 2)$ can be priced without ambiguity.

5 Confrontation with data

The construction has one dimensionful input and, at tree level, one output ratio; the on-shell relation $\sin^2 \theta_W = 1 - m_W^2/m_Z^2$ makes the m_W comparison and the on-shell-angle comparison the *same* test, and we present it once, as a prediction for m_W given m_Z :

$$M_\infty = \frac{m_Z}{\sqrt{\lambda_+(2)}} = m_Z \sqrt{\frac{1 + \sqrt{3}}{2}}, \quad m_W^{\text{pred}} = m_Z \sqrt{\rho}. \quad (11)$$

With $m_Z = 91.1876(21)$ GeV [20],

$$M_\infty = 106.578 \text{ GeV}, \quad m_W^{\text{pred}} = 80.3745(19) \text{ GeV}, \quad (12)$$

where the quoted uncertainty is input propagation only ($\partial m_W / \partial m_Z = \sqrt{\rho}$). Against the measured values (pull $\equiv (m_W^{\text{pred}} - m_W^{\text{meas}}) / \sigma_{\text{comb}}$, combining the experimental error with the ± 0.0019 input error in quadrature):

measurement	m_W [GeV]	source	pull
PDG 2024 world average	80.3692(133)	[20]	+0.4 σ
CMS 2024	80.3602(99)	[21]	+1.4 σ
PDG-live 2026 snapshot	80.3625(77)	[corpus pin, unverified]	+1.5 σ
CDF II 2022	80.4335(94)	[22]	−6.1 σ

Equivalently, the predicted on-shell angle is $\sin^2 \theta_W = 0.2231013$, to be compared with 0.22305(23) [20] (+0.2 σ); the corpus also pins an unverified 2026 snapshot 0.223339(153) (−1.6 σ). Three honesty notes are mandatory.

(a) *Scheme.* The algebraic prediction is a pure number; the data are scheme- and scale-dependent. The comparison above uses the on-shell definition, for which $\sin^2 \theta_W \equiv 1 - m_W^2/m_Z^2$ is an identity. The effective leptonic angle $\sin^2 \theta_{\text{eff}}^\ell \simeq 0.2315$ differs from the on-shell value by known radiative corrections of order 4%; even the mass *definition* matters at this precision, since the conventional Breit–Wigner (running-width) masses and the complex-pole masses differ by $\Gamma^2/2m$, roughly 34 MeV for the Z and 27 MeV for the W — an order of magnitude above the experimental errors — and the construction has not declared which definition it predicts. A claim that the kernel value targets one scheme and one mass definition rather than another is dynamical input the construction does not yet possess. There is also a sharper internal tension: the SM global electroweak fit returns an indirect $m_W \simeq 80.355(6)$ GeV [20] from m_Z , m_t , m_H and the couplings via the radiative Δr relation, and the algebraic value 80.3745 lies about 3 σ *above* it; if the on-shell prediction were exact, consistency would require a compensating new-physics contribution to Δr — an unpleasant but usefully concrete corollary. The necessary next step is a one-loop matching computation determining whether any consistent scheme-and-definition assignment survives at current precision.

(b) *Data motion.* The pull moved from +0.4 σ (2024 world average) to about +1.5 σ (CMS 2024 and the 2026 snapshot) while CDF II sits 6 σ on the other side; the experimental situation for

m_W is unsettled. A retrodiction study across PDG editions — checking whether the agreement improved as errors shrank, the signature of a real constraint rather than a fit to current centrals — is defined in the corpus’s planning documents and should be executed before stronger claims are made.

(c) *Error budget.* The theory error (scheme choice, radiative corrections, and the unknown dynamics behind the kernel) dominates the ± 0.0019 input error and is presently unquantified. Any use of (12) as a precision claim without that budget would be overreach.

6 The positive-branch tower

If assignment (4) extends to higher j , the positive branch is a tower of levels below M_∞ :

$$m(j) = M_\infty \sqrt{\lambda_+(j(j+1))}, \quad \lambda_+(J) = 1 - \frac{1}{J} + \mathcal{O}\left(\frac{1}{J^2}\right), \quad m(j) \simeq M_\infty \left(1 - \frac{1}{2J}\right) \quad (J \rightarrow \infty). \quad (13)$$

Exact closed forms and hand-evaluated values:

j	J	$\lambda_+(J)$	$m(j)$
$\frac{1}{2}$	$\frac{3}{4}$	$(\sqrt{57} - 3)/8 = 0.568729$	80.374 GeV ($= m_W^{\text{pred}}$)
1	2	$\sqrt{3} - 1 = 0.732051$	91.188 GeV ($= m_Z$, input)
$\frac{3}{2}$	$\frac{15}{4}$	$(\sqrt{465} - 15)/8 = 0.820482$	96.54 GeV
2	6	$(\sqrt{60} - 6)/2 = 0.872983$	99.58 GeV
$\frac{5}{2}$	$\frac{35}{4}$	$(\sqrt{1785} - 35)/8 = 0.906157$	101.45 GeV
3	12	$(\sqrt{192} - 12)/2 = 0.928203$	102.68 GeV
$\vdots$			accumulating at $M_\infty = 106.58$ GeV

The accumulation point carries a consistency obligation the source corpus does not discuss: taken literally, (13) places infinitely many levels below 107 GeV, with spacing $\sim M_\infty/2J$. Compatibility with LEP and electroweak precision data then *requires* the couplings of the levels to SM fields to fall rapidly with j — which is exactly what the label-conservation pattern of the next subsection supplies, since reaching a level of internal label j from SM states ($j \leq 1$) demands increasingly label-violating operators. Either the tower terminates dynamically, or its visibility must decay with j ; a derivation must produce one of the two, and precision Z -pole and LEP 2 data already bound the alternatives.

6.1 A label-conservation pattern and the 95–96 GeV hints

The following discussion is conditional on Assumption 1 below (approximate conservation of the internal label by the production and decay dynamics) and is qualitative: the 4D spin, CP, and width of the tower states are unspecified by the kernel, and rate predictions require the label-breaking dynamics, which the construction does not yet possess. What the label bookkeeping alone gives is a *pattern*.

With the SM bosons carrying $j_\gamma = 0$, $j_W = \frac{1}{2}$, $j_Z = 1$, two-boson channels carry

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1, \quad 1 \otimes 1 = 0 \oplus 1 \oplus 2, \quad \frac{1}{2} \otimes 1 = \frac{1}{2} \oplus \frac{3}{2}, \quad 1 \otimes 0 = 1, \quad \frac{1}{2} \otimes 0 = \frac{1}{2}.$$

Consequences for the electrically neutral tower members:

- The $j = \frac{3}{2}$ level at 96.54 GeV appears in no neutral two-boson channel: $\frac{3}{2} \not\subset 0 \oplus 1$ (WW , $Z\gamma$, $\gamma\gamma$) and $\frac{3}{2} \not\subset 0 \oplus 1 \oplus 2$ (ZZ). The only two-boson channel containing $\frac{3}{2}$ is $W^\pm Z$, which is electrically charged and available only to its charged partners. Label-conserving single production of the neutral $j = \frac{3}{2}$ state in Drell–Yan or vector-boson fusion is therefore forbidden; it proceeds only through label breaking or higher-dimensional operators, and $\gamma\gamma$

couplings arise at loop level. A narrow, weakly produced state near 96.5 GeV is *compatible* with — though in no way established by — the unresolved low-mass diphoton hints near 95–96 GeV (CMS [23]: local 2.9σ at 95.4 GeV; ATLAS [24]: local 1.7σ at the same mass). The pattern statement is falsifiable in shape: within this construction such a state should remain weak in single production, while pair production opens at $\sqrt{s} > 2m(3/2) = 193$ GeV.

- The $j = 2$ level at 99.58 GeV does occur in $1 \otimes 1$: it is the first tower state with a label-allowed two- Z vertex, so it would appear in ZZ^* and near-threshold ZZ observables rather than in $\gamma\gamma$.

Two data-driven tests follow. LEP 2 accumulated data up to $\sqrt{s} = 209$ GeV, above the $j = 3/2$ pair threshold of 193 GeV: a dedicated recast of LEP 2 cross sections for pair-produced narrow neutral states in the 96–104 GeV range would already constrain the tower and is, to our knowledge, the sharpest retrodictive test available. FCC-ee, with runs at 240 GeV and precision scans, either finds the tower or excludes it up to the accumulation point 106.58 GeV.

7 The negative branch: sum rules and a no-go

The corpus matches the negative-branch levels to the electroweak vacuum expectation value $v = 246.220$ GeV:

$$m_{-}(j) \equiv M_{\infty} \sqrt{|\lambda_{-}(j(j+1))|} : \quad m_{-}(\tfrac{1}{2}) = 122.39 \text{ GeV} \leftrightarrow \tfrac{v}{2} = 123.11 \text{ GeV}, \quad m_{-}(1) = 176.16 \text{ GeV} \leftrightarrow \tfrac{v}{\sqrt{2}} = \quad (14)$$

with deviations -0.58% and $+1.18\%$ respectively. The $J = 2$ identification has a strikingly compact exact form worth recording: since $|\lambda_{-}(2)|/\lambda_{+}(2) = (1 + \sqrt{3})/(\sqrt{3} - 1) = (1 + \sqrt{3})^2/2$, it is equivalent to

$$v = (1 + \sqrt{3}) m_Z = |\lambda_{-}(2)| m_Z = 249.13 \text{ GeV} \quad (\text{vs. } 246.22, +1.18\%), \quad (15)$$

the vacuum expectation value as the negative-branch *eigenvalue* itself in units of m_Z . We emphasize the epistemic status: pending a mechanism that interprets the negative branch, the identifications (14) are numerology; what this section contributes is the exact algebra they must satisfy, and a no-go they currently violate.

From (5), with no further input:

Proposition 2 (Branch sum rules). *For every j , with $J = j(j+1)$,*

$$m_{-}^2(j) - m_{+}^2(j) = J M_{\infty}^2, \quad m_{+}(j) m_{-}(j) = \sqrt{J} M_{\infty}^2.$$

Inserting the identifications (14) and the measured m_W, m_Z , the first rule holds at $+2.1\%$ ($J = \frac{3}{4}$) and -3.2% ($J = 2$); the second at $+0.6\%$ and -1.2% — i.e. exactly at the level of the mass matches themselves, as must happen algebraically.

More constraining is that the two identifications fight each other:

Lemma 1 (The two vacuum identifications cannot both be exact). *If $m_{-}(\frac{1}{2}) = v/2$ and $m_{-}(1) = v/\sqrt{2}$ held simultaneously and exactly, then*

$$\frac{|\lambda_{-}(2)|}{|\lambda_{-}(3/4)|} = \frac{(v/\sqrt{2})^2}{(v/2)^2} = 2.$$

But the exact value is

$$\frac{|\lambda_{-}(2)|}{|\lambda_{-}(3/4)|} = \frac{8(1 + \sqrt{3})}{3 + \sqrt{57}} = 2.07173 \dots,$$

which differs from 2 by 3.6% (i.e. 1.8% in the mass ratio — realized in the data as the observed $-0.58\%/+1.18\%$ split). Hence at most one of the two identifications can be exact; the pair is either approximate (radiatively corrected) or partially accidental.

This no-go, which to our knowledge is stated here for the first time, sharpens the negative-branch question considerably: any future derivation must either pick one exact identification and explain the other’s percent-level offset, or predict the specific combination $8(1 + \sqrt{3})/(3 + \sqrt{57})$ in place of 2 — e.g. the conventional ratio $(v/\sqrt{2})/(v/2) = \sqrt{2} = 1.41421$ would be replaced by $\sqrt{2.07173} = 1.43935$.

8 The Casimir coincidence $\kappa = 3/8$

A custodial-EFT analysis in the corpus, working from the same electroweak data with entirely different methods (Wilson-coefficient fits plus a look-elsewhere analysis), isolated the coefficient

$$\kappa = \frac{3}{8} = \frac{C_F}{C_A} \Big|_{SU(2)}$$

as the group-theoretic content of its best-fit relation. We record that this is the same pair of Casimirs that powers the kernel:

$$\frac{J_W}{J_Z} = \frac{C_2(\text{fund})}{C_2(\text{adj})} = \frac{3/4}{2} = \frac{3}{8}.$$

Theorem 2 makes the cross-link exact rather than suggestive: κ is literally a coordinate of the quartic family, entering its two lowest coefficients as $e_3 = e_1(1 - \kappa)$ and $e_4 = (1 - \kappa)^2$, so “the quartic matches data” and “ $\kappa = 3/8$ ” are two readings of one algebraic object evaluated at $(I, \kappa) = (\frac{3}{4}, \frac{3}{8})$. Given Theorem 1, three formulations — kernel, quartic, custodial coefficient — are now built on the single Casimir pair $(\frac{3}{4}, 2)$. Whether this is over-determination or one coincidence wearing three coats is exactly the question a trials-corrected analysis must answer; the equivalence results of this paper are necessary bookkeeping input to that computation, since they prevent double counting.

9 Toward a geometric origin

Nothing above is a dynamical theory: (1) is a spectral ansatz. We now state what a derivation must produce, and why Kaluza–Klein-type geometry is the natural habitat, without committing to a specific internal manifold.

Assumption 1 (Internal label). There is an internal $SU(2)_i$ — an isometry of an internal space or a custodial remnant — under which the electroweak bosons carry $(\gamma, W, Z) = (0, \frac{1}{2}, 1)$, and which is approximately conserved by the dynamics that builds the tower.

Assumption 2 (Pairing through the generators). The visible state $|v\rangle$ of the spin- j multiplet couples to its partner states through the internal generators $\vec{J}$ themselves, with one universal normalization: the coupling row vector is $\langle p | \vec{J} | v \rangle$ over partner states $|p\rangle$. Its squared norm is

$$\sum_p |\langle p | \vec{J} | v \rangle|^2 = \langle v | \vec{J}^2 | v \rangle = j(j+1) = J$$

for *every* state of *every* representation, so the two-block reduction onto $(|v\rangle, \vec{J}|v\rangle/\sqrt{J})$ has off-diagonal exactly $\sqrt{J}$, uniformly in j , integer or half-integer, and independently of any basis choice. (A per-component version — coupling through a single ladder matrix element $\langle j, m \pm 1 | J_{\pm} | j, m \rangle$ — reproduces $\sqrt{J}$ only for integer j from the $m = 0$ state and fails for half-integer j ; the operator-norm version above supersedes it.)

Assumption 3 (Partner mass, sign, and nesting). The partner multiplet carries diagonal mass-squared J in the same units — the Casimir eigenvalue of an internal Laplacian, as on any coset

space, where the relevant spectra are universal multiples of C_2 [16, 19] — with the tachyonic sign and the nested-decimation structure of Proposition 1(iii), for which curvature-induced negative vector-mode directions on distorted internal spaces [17, 19] and Wilson-line (Hosotani) dynamics [18] are the known geometric sources.

Under Assumptions 1–3 the spectrum (3) follows. Assumption 2 is the least negotiable ingredient and also the most suggestive: the equality of the coupling with $\sqrt{C_2}$ — the corpus’s “coefficient lock” — holds automatically when the mixing operator *is* the generator, normalized by the same constant that normalizes the Casimir. That is precisely the structure of a gauge coupling to isometry currents, which is why we regard Kaluza–Klein geometry [13, 14, 15, 16] as the natural habitat: any mechanism producing instead a mixing $g\vec{J}$ with an independent strength $g \neq 1$ is incompatible with the observed five-digit agreement of m_W/m_Z . What no calculation in the corpus or, to our knowledge, in the literature has yet produced is the conjunction of all three assumptions in one geometry — with Remark 1 now adding a structural constraint: the geometry must realize a nested decimation, and flat-chain constructions are excluded. The corpus’s classification-sweep program (enumerating coset geometries and testing which, if any, produce the lock) is the right instrument, and its outcome — a derivation or a clean no-go — will decide whether this section describes physics or bookkeeping.

10 Failure conditions

The construction is falsifiable on several independent fronts.

1. *m_W precision.* The tree-level on-shell prediction is $m_W = 80.3745(19)$ GeV given m_Z . A settled world average more than $\sim 2\sigma$ away kills the positive branch as an exact on-shell statement; the 2024–2026 trend (CMS at $+1.4\sigma$, the 2026 snapshot at $+1.5\sigma$) is mildly unfavorable and moving, while any confirmation of the CDF II region above 80.40 GeV excludes it outright.
2. *Scheme stability.* If the one-loop matching program shows that no consistent scheme assignment reproduces the algebraic number within errors, the five-digit match is an on-shell accident.
3. *The tower.* LEP 2 recasts above 193 GeV and FCC-ee either find neutral states at 96.5/99.6/101.5/102.7 GeV with the predicted weak-single/allowed-pair pattern, or exclude the tower up to 106.58 GeV, falsifying the extended assignment outright.
4. *Negative branch.* Lemma 1 forces a choice; independent precision on v , m_t , and the Higgs potential pins both conventional scales, and the surviving identification then predicts the other’s offset (1.8% in ratio) with no free parameter.
5. *Statistics.* A trials-corrected analysis over the full choice space (Section 11) may return an unremarkable probability — the soft but decisive failure mode for the entire enterprise.

11 Method, provenance, and trials accounting

This manuscript was produced by a language model working over a corpus of research repositories, under a constraint of zero code execution: every closed form was derived symbolically in the model’s working notes, and every decimal was obtained by hand arithmetic from those closed forms, rounded at the last digit displayed. The constraint delimits the capability being demonstrated: mathematical synthesis — noticing that two formulations are one (Theorem 1), that a truncation is secretly a resummation with a spectral obstruction (Proposition 1, Remark 1), that two identifications are mutually inconsistent (Lemma 1) — rather than numerical search. An adversarial referee round by an independent model materially improved the paper: it caught a

false identity in the first draft’s irreducibility argument (since replaced by the orbit argument in Theorem 1), forced the honest reframing of Proposition 1, and corrected a mis-stated experimental pull.

Provenance, stated exactly:

- *Inherited*: the empirical observation and channel assignment (4) [1, 2, 3]; the secular equation and its two-channel reading, with the mixing value 0.22310132 stated already in [3]; the quartic (10), the negative-branch identifications (14), and the $\kappa = 3/8$ EFT result, all from the unpublished corpus; the experimental inputs [20, 21, 22].
- *Reproved from scratch*: the closed spectrum (Section 3); the convergence of the decimation recursion.
- *New here, to our knowledge*: the proof of Theorem 1 (the corpus had verified the coincidence numerically, never algebraically); the identification of the quartic’s four roots with the four branch combinations and the two-parameter family of Theorem 2, whose coefficients are polynomial in (I, κ) and which hands the statistics program its natural hypothesis class; Corollary 1 with the degree-8 proof; the flat-chain obstruction of Remark 1, including the threshold $J = \frac{9}{4}$; the operator-norm formulation of the coefficient lock in Assumption 2, which repairs the half-integer case; the branch sum rules in linking form (Proposition 2); the no-go of Lemma 1 with its 1.8% figure and the compact reformulation $v = (1 + \sqrt{3}) m_Z$ of the $J=2$ identification (15); the label-conservation pattern of Section 6.1 with the LEP 2 recast and the 193 GeV pair threshold; and the observation that the biquadratic field makes $\sin^2 \theta_W$ ruler-and-compass constructible.

Trials accounting. Any significance claim for this construction must price, at minimum, the following post-hoc choices: the kernel family (1) among comparably simple 2×2 textures; the map $J = j(j+1)$ rather than another Casimir function; the positive-branch selection; the channel assignment (4); the anchor at m_Z ; the on-shell scheme; the extension of the tower and the association with the 95–96 GeV hints; and the negative-branch matches (14). The corpus contains Monte Carlo look-elsewhere machinery and quotes trials-corrected probabilities of order 10^{-4} – 10^{-6} for parts of this structure; those numbers are unaudited here and no significance is claimed on their basis. What this paper contributes to that future audit is subtractive and additive at once: Theorem 1 collapses two formulations into one (reducing the apparent confirmation count), while Lemma 1 shows the negative-branch sector currently *fails* an exactness test — both facts that an honest significance computation must incorporate.

The honest prior on constructions of this genre is unfavorable: the history of electroweak numerology is a graveyard, and the Koide relation itself, after forty years of real theoretical effort [6, 10, 11], remains a suggestive orphan. The construction studied here has three properties that graveyard occupants typically lack: a five-digit match with one input and no continuous parameters; an internal algebraic rigidity that generates new exact statements when probed (Sections 4, 7); and a finite list of near-term experimental executioners (Section 10). We submit that this is the correct standard for work of this kind: a well-posed falsification program.

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